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Work machine

Abstract

A work machine including a vehicle body configured to travel, a fan, an obstacle detection device, and a controller. The fan is configured to exchange air between an inside and an outside of the vehicle body through an opening provided toward a rear of the vehicle body. The obstacle detection device is installed behind the fan. The obstacle detection device is configured to detect an obstacle in the rear of the vehicle body. The controller is configured to control intake and exhaust of the fan based on a state of the obstacle detection device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is a U.S. National stage application of International Application No. PCT/JP2021/027034, filed on Jul. 19, 2021. This U.S. National stage application claims priority under 35 U.S.C. § 119(a) to Japanese Patent Application No. 2020-165735, filed in Japan on Sep. 30, 2020, the entire contents of which are hereby incorporated herein by reference.
BACKGROUND
Technical Field
(2) The present invention relates to a work machine.
Background Information
(3) Collision suppression system has been proposed for detecting an obstacle to the rear and suppressing a collision in a work machine such as a wheel loader (for example, see Registration of Utility Model No. 3219005).
(4) In such a collision suppression system, a detection device such as a radar is provided at the rear end of the work machine.
(5) Further, in the wheel loader, an engine room is provided at the rear of the vehicle. A fan for cooling the engine cooling water is disposed in the rear part of the engine room. By driving the fan, air is taken in or exhausted between the inside and outside of the engine room through an opening formed at the rear end of the engine room.
SUMMARY
(6) However, when the detection device is installed on the rear end side of the vehicle relative to the fan, for example, in the case of being configured to cool at the time of exhaust, high-temperature air warmed inside the engine room hits the detection device. As a result, the temperature of the detection device may exceed the operating range and the detection performance of the detection device may deteriorate. In addition, in the case of being configured to cool at the time of intake, external dust or the like may adhere to the surface of the detection device, deteriorating the detection performance.
(7) An object of the present disclosure is to provide a work machine capable of suppressing deterioration in performance for detecting an obstacle in the rear.
(8) A work machine according to a disclosure includes a vehicle body configured to travel, a fan, an obstacle detection device, and a controller. The fan exchanges air between an inside and an outside of the vehicle body through an opening provided toward a rear of the vehicle body. The obstacle detection device is installed behind the cooling device and detects an obstacle in the rear of the vehicle body. The controller controls intake and exhaust of the fan based on a state of the obstacle detection device.
(9) According to the present disclosure, it is possible to provide a work machine capable of suppressing deterioration in the performance of detecting an obstacle in the rear.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a side view of a wheel loader of a first embodiment according to the present disclosure.
(2) FIG. **2**A is a perspective view illustrating the rear part of the wheel loader of the first embodiment according to the present disclosure, FIG. **2**B is an enlarged view of the radar device of FIG. **2**A, and FIG. **2**C is a view illustrating a state in which a cover is removed from FIG. **2**B.
(3) FIG. **3** is a side view illustrating the rear part of the wheel loader of the first embodiment according to the present disclosure.
(4) FIG. **4** is a diagram illustrating configurations of a cooling device and a control system in the wheel loader of the first embodiment according to the present disclosure.

(5) FIG. **5** is a flowchart illustrating control of the wheel loader of the first embodiment according to the present disclosure.
(6) FIG. **6** is a perspective view illustrating the rear part in the wheel loader of a second embodiment according to the present disclosure.
(7) FIG. **7** is a diagram illustrating configurations of a cooling device and a control system in the wheel loader of the second embodiment according to the present disclosure.
(8) FIG. **8** is a flowchart illustrating control of the wheel loader of the second embodiment according to the present disclosure.
(9) FIG. **9** is a diagram illustrating configurations of a cooling device and a control system in the wheel loader of a modified example of the first embodiment according to the present disclosure.
(10) FIG. **10** is a diagram illustrating configurations of a cooling device and a control system in the wheel loader of a modified example of the second embodiment according to the present disclosure.
DESCRIPTION OF EMBODIMENTS
(11) A wheel loader as an example of a work machine according to the present disclosure will be described below with reference to the drawings.
Embodiment 1
(12) (Outline of Wheel Loader)
(13) FIG. **1** is a side view illustrating the wheel loader **100** (an example of a work machine) of present embodiment.
(14) The wheel loader **100** of the present embodiment includes a vehicle body **1**, a cooling device **30** (see FIG. **4**), a radar device **40** (an example of an obstacle detection device), a temperature sensor **50**, and a controller **60** (see FIG. **4**).
(15) The vehicle body **1** is able to travel. The cooling device **30** cools the cooling water of the engine of the vehicle body **1** and the like. The radar device **40** detects an obstacle in a rear of the vehicle body **1**. A temperature sensor **50** measures the temperature of the radar device **40**. The controller **60** controls intake and exhaust of the cooling device **30** based on the detected value of the temperature sensor **50**.
(16) A vehicle body **1** includes a vehicle body frame **2**, a work implement **3**, a pair of front tires **4** (an example of traveling wheels), a cab **5**, an engine room **6**, a pair of rear tires **7** (an example of traveling wheels), a steering cylinder **9**, and a counterweight **10**.
(17) In the following description, “front”, “rear”, “right”, “left”, “up”, and “down” indicate directions relative to a state of looking forward from the driver's seat. Also, “vehicle width direction” and “left-right direction” have the same meaning. In FIG. **1**, the front-rear direction is indicated by X, the front direction is indicated by Xf, and the rear direction is indicated by Xb.
(18) The wheel loader **100** performs work such as earth and sand loading by using the work implement **3**.
(19) The vehicle body frame **2** is a so-called articulated construction and includes a front frame **11**, a rear frame **12**, and a coupling shaft part **13**. The front frame **11** is disposed in front of the rear frame **12**. The coupling shaft part **13** is provided in the center in the vehicle width direction and couples the front frame **11** and the rear frame **12** to each other in a manner that allows swinging. The pair of front tires **4** are attached to the left and right of the front frame **11**. The pair of rear tires **7** are attached to the left and right of the rear frame **12**.
(20) The work implement **3** is driven by hydraulic fluid from a work implement pump (not shown). The work implement **3** is attached to the front part of the front frame **11** in a manner that allows swinging. The work implement **3** includes a boom **14**, a bucket **15**, a lift cylinder **16**, a bucket cylinder **17**, and a bell crank **18**.
(21) The base end of the boom **14** is rotatably attached to a front part of the front frame **11**. The tip end of the boom **14** is rotatably attached to a rear part of the bucket **15**. The rear part of the bucket **15** is on the opposite side from an opening **15***b*. The tip end of a cylinder rod of the lift cylinder **16** is rotatably attached between the base end and the tip end of the boom **14**. The cylinder body of the

lift cylinder **16** is rotatably attached to the front frame **11**.
(22) One end part of the bell crank **18** is rotatably attached to the tip end of a cylinder rod of the bucket cylinder **17**. The other end part of the bell crank **18** is rotatably attached to a rear part of the bucket **15**. The bell crank **18** is rotatably supported by a bell crank support **14***d* near the middle of the boom **14** between either end. The cylinder body of the bucket cylinder **17** is rotatably attached to the front frame **11**. The extension/contraction force of the bucket cylinder **17** is converted to rotational movement by the bell crank **18** and is transmitted to the bucket **15**.
(23) The bucket **15** is rotatably attached to the tip end of the boom **14** so as to open toward the front. By extending and contracting the bucket cylinder **17**, the bucket **15** rotates with respect to the boom **14** and performs a tilting motion and a dumping motion.
(24) The cab **5** is disposed on the rear frame **12**. A steering handle for performing steering operations, a lever for operating the work implement **3**, and various display devices are disposed inside the cab **5**. The engine room **6** is disposed to the rear of the cab **5** and on the rear frame **12** and accommodates an engine (not shown).
(25) The counterweight **10** is provided at the rear part of the rear frame **12**. The counterweight **10** is disposed below the engine room **6**. A part of the counterweight **10** is located in the rear of the engine room **6**. A rear end **10***e* which is an end of the counterweight **10** on the rear direction Xb side corresponds to a rear end of the wheel loader **100**.
(26) (Cooling Device **30**)
(27) FIG. **2**A is a perspective view illustrating a rear part of the vehicle body **1**. FIG. **3** is a side view illustrating the rear part of the vehicle body **1**. As shown in FIG. **3**, an engine **19**, a radiator **20**, and a fan **31** of a cooling device **30** are disposed in order from the front along the front-rear direction X inside the engine room C.
(28) The engine room **6** includes a main body part **6***a* and a grille **6***b*. The main body part **6***a* is provided so as to form side plates that covers the front and left and right sides of the engine **19** and the radiator **20**, and to form a top plate that covers the top of the engine **19** and the radiator **20**.
(29) The grille **6***b* is disposed at a rear end of the main body part Ca. The grille **6***b* is formed so as to cover the fan **31** in general. A grid-like opening **6***c* is formed in a rear surface of the grille **6***b*. Intake and exhaust are performed between the inside and the outside of the engine room **6** through the opening **6***c*. Although not shown, the grille **6***b* is rotatable upward or laterally with respect to the main body part **6***a*, so that the inside of the engine room **6** can be cleaned.
(30) The radiator **20** cools, for example, cooling water for cooling the engine **19** by exchanging heat with air supplied by driving the fan **31**.
(31) In the cooling device **30** of the present embodiment, the fan **31** is disposed downstream of the radiator **20** which is an object to be cooled, and the fan **31** rotates to cool the radiator **20** in an exhaust state (see arrow B in FIG. **3**). The exhaust state is a state in which air is discharged from the inside of the engine room **6** to the outside through the opening **6***c*. The intake state is a state in which air is sucked from the outside of the engine room **6** toward the inside through the opening **6***c*.
(32) Next, the cooling device **30** that is able to switch the rotation of the fan **31** between the intake state and the exhaust state will be described.
(33) FIG. **4** is a view illustrating configurations of the cooling device **30** and the control system. As shown in FIG. **4**, the cooling device **30** includes the fan **31** described above, a hydraulic pump **32** (an example of a pump), a hydraulic fluid tank **33**, a hydraulic motor **34** (an example of a motor), a first passage **35**., a second passage **36**, and a directional control valve **37**.
(34) The hydraulic pump **32** is rotationally driven by the shaft output of the engine **19**. The hydraulic pump **32** sucks hydraulic fluid from the hydraulic fluid tank **33**, discharges high pressure oil, and supplies the hydraulic fluid to the hydraulic motor **34**.
(35) The hydraulic motor **34** is driven by the hydraulic pressure supplied by the hydraulic pump **32** and rotates the fan **31** with shaft rotation output.

(36) The first passage **35** and the second passage **36** connect the hydraulic pump **32** and the hydraulic fluid tank **33** to the hydraulic motor **34**. The first passage **35** and the second passage **36** supply pressure oil supplied from the hydraulic pump **32** to the hydraulic motor **34** or discharge pressure oil supplied from the hydraulic pump **32** from the hydraulic motor **34**.
(37) The directional control valve **37** is provided in the middle of the first passage **35** and in the middle of the second passage **36**. A portion of the first passage **35** from the hydraulic pump **32** to the directional control valve **37** is referred to as a passage portion **35***a*, and a portion of the first passage **35** from the directional control valve **37** to the hydraulic motor **34** is referred to as a passage portion **35***b*. A portion of the second passage **36** from the hydraulic fluid tank **33** to the directional control valve **37** is defined as a passage portion **36***a*, and a portion of the second passage **36** from the directional control valve **37** to the hydraulic motor **34** is defined as a passage portion **36***b*.
(38) The directional control valve **37** is, for example, a 4-port 3-position directional control valve. The directional control valve **37** is a solenoid valve. The directional control valve **37** includes a valve body, and is able to switch so that the valve body is positioned at any one of a stop position P**0**, a first position P**1**, or a second position P**2** based on a command signal from the controller **60**. At the stop position P**0**, the supply and discharge of pressure oil are stopped, and the rotation of the hydraulic motor **34** is stopped.
(39) When the directional control valve **37** is switched from the stop position P**0** to the first position P**1**, the hydraulic fluid is supplied from the hydraulic pump **32** to the passage portion **35***a*, the passage portion **35***b*, the hydraulic motor **34**, the passage portion **36***b*, the passage portion **36***a*, and the hydraulic fluid tank **33** in this order, and the hydraulic motor **34** rotates in the normal direction. On the other hand, when the directional control valve **37** is switched from the first position P**1** to the second position P**2**, the hydraulic fluid is supplied from the hydraulic pump **32** to the passage portion **35***a*, the passage portion **36***b*, the hydraulic motor **34**, the passage portion **35***b*, the passage portion **36***a*, and to the hydraulic fluid tank **33** in this order, and the hydraulic motor **34** rotates in the reverse direction.
(40) In the present embodiment, when the hydraulic motor **34** rotates in the normal direction, the fan **31** rotates in the normal direction to enter the exhaust state. Further, when the hydraulic motor **34** rotates in the reverse direction, the fan **31** rotates in the reverse direction to enter the intake state.
(41) That is, by driving the hydraulic pump **32** in a state in which the directional control valve **37** is switched to the first position P**1**, the fan **31** rotates in the normal direction to enter the exhaust state. Further, by driving the hydraulic pump **32** in a state in which the directional control valve **37** is switched to the second position P**2**, the fan **31** rotates in the reverse direction to enter the intake state.
(42) (Radar Device **40**)
(43) The radar device **40** detects the presence of an obstacle S in the rear of the vehicle body **1**, as shown in FIG. **1**. In this embodiment, for example, as shown in FIG. **2**A, two radar devices **40** are provided. One radar device **40** is provided on the left side of the central axis A in the left-right direction of the rear frame **12**, and the other radar device **40** is provided on the right side of the central axis A. The two radar devices **40** are disposed symmetrically with the central axis A interposed therebetween. The number of radar devices **40** is not particularly limited. For example, one radar device **40** may be provided on the central axis A of the rear frame **12**, or three or more may be provided.
(44) The radar devices **40** are disposed above the counterweight **10**. As shown in FIG. **2**A and FIG. **3**, the radar device **40** is disposed below the grille **6***b* and on the rear direction Xb side of the grille **6***b*. The radar devices **40** are fixed to the grille **6***b*.
(45) The radar device **40** is, for example, a millimeter wave radar. The radar device **40** can measure the distance to an object by detecting, with the receiving antenna, the radio waves in the millimeter

wave band which is emitted from the transmitting antenna and is returned after being reflected by the surface of the obstacle. A detection result by the radar device **40** is transmitted to a processor (not shown), and the processor can detect that an obstacle exists within a predetermined range when the wheel loader **100** is traveling in the rear direction.

(46) FIG. **2**B is a perspective view of the radar device **40** as seen from the rear. FIG. **2**C is a view illustrating a state in which the cover part **41***b* is removed from the configuration shown in FIG. **2**B. The radar device **40** includes, for example, a housing **41** and a detector **42**. The housing **41** accommodates detector **42**. The housing **41** includes a support part **41***a* that supports the detector **42** and a cover part **41***b* that covers a surface of the detector **42**. A support part **41***a* is fixed to the grille **6***b*. As shown in FIG. **2**B, the detector **42** is fixed to a surface of the support part **41***a* of the housing **41**. The detector **42** is provided with a transmit antenna and a receive antenna. The detector **42** is attached to the support part **41***a* so that the detection surface **42***a* faces to the rear direction.

(47) (Temperature Sensor **50**)

(48) The temperature sensor **50** is disposed inside the housing **41** of the radar device **40**, for example, as shown in FIG. **2**C. The temperature sensor **50** measures the temperature inside the housing **41** of the radar device **40**, but is not limited to this, and may measure the surface temperature of the radar device **40** (the detection surface **42***a* and the cover part **42***b*) in a non-contact manner.

(49) When a plurality of radar devices **40** are provided as shown in FIG. **2**B, the temperature sensor **50** may be provided in each radar device **40** or only in some of the radar devices.

(50) A value detected by the temperature sensor **50** is transmitted to the controller **60** as a detection signal as shown in FIG. **4** which will be described later.

(51) (Controller **60**)

(52) The controller **60** includes a processor and a storage device. The processor is, for example, a central processing unit (CPU). Alternatively, the processor may be a processor different from a CPU. The processor executes processing for controlling the wheel loader **100** in accordance with a program. The storage device includes a non-volatile memory such as a read-only memory (ROM) and a volatile memory such as a random access memory (RAM). The storage device may include an auxiliary storage device such as a hard disk or a solid state drive (SSD). The storage device is an example of a non-transitory computer-readable recording medium. The storage device stores data and a program for controlling the wheel loader **100**. The storage device stores, for example, the data of belowmentioned thresholds.

(53) The controller **60** normally controls the directional control valve **37** so that the valve body is positioned at the first position P**1**, and rotates the fan **31** in the normal direction to cool the radiator **20**.

(54) A value detected by the temperature sensor **50** is input to the controller **60** as a detection signal. The controller **60** stores a temperature threshold value, and when the value detected by the temperature sensor **50** exceeds the threshold value, the controller **60** controls the directional control valve **37** so that the valve body is positioned at the second position P**2**, and rotates the fan **31** in the reverse direction.

(55) In this way, by rotating the fan **31** in the reverse direction, the fan **31** enters to the intake state and air from the outside hits the radar device **40** to cool the radar device **40**, so it is possible to lower the temperature.

(56) Although only one threshold may be used, the temperature threshold for switching from the exhaust state to the intake state and the temperature threshold for returning from the intake state to the exhaust state may be changed. In this case, it is preferable to lower the second threshold value of the temperature for returning to the exhaust state rather than the first threshold value of the temperature for changing to the intake state.

(57) (Actions)

(58) Next, control of wheel loader **100** in the present embodiment will be described. FIG. **5** is a

flowchart illustrating control of wheel loader **100** of the present embodiment.
(59) FIG. **5** shows a case where the second threshold value of the temperature when returning to the exhaust state is set lower than the first threshold value of the temperature when changing to the intake state.
(60) First, in step S**10**, when the engine **19** of the wheel loader **100** is started, the controller **60** sets the directional control valve **37** to a state in which the valve body is disposed at the first position P**1**, drives the hydraulic pump **32** to rotate the fan **31** in the normal direction, and sets the fan **31** to the exhaust state.
(61) Next, in step S**20**, the controller **60** determines whether the temperature of the radar device **40** detected by the temperature sensor **50** is equal to or higher than the first threshold. When the detected temperature is less than the first threshold, the control in step S**20** is repeated to keep the fan **31** rotating in the normal direction.
(62) On the other hand, when it is determined in step S**20** that the temperature detected by the temperature sensor **50** is equal to or higher than the first threshold, the controller **60** switches the directional control valve **37** from the first position P**1** to the second position P**2** in step S**30**, rotate the fan **31** in the reverse direction, and sets the fan **31** to the intake state.
(63) Next, in step S**40**, the controller **60** determines whether or not the temperature detected by the temperature sensor **50** is equal to or lower than the second threshold. When the detected temperature is higher than the second threshold, the control in step S**40** is repeated to keep the fan **31** rotating in the reverse direction.
(64) On the other hand, when it is determined in step S**40** that the temperature detected by the temperature sensor **50** is equal to or lower than the second threshold, the controller **60** switches the directional control valve **37** from the second position P**2** to the first position P**1** in step S**50**, rotates the fan **31** in the normal direction, and sets the fan **31** to the exhaust state. After step S**50**, the control returns to step S**20**, and steps S**20** to S**50** are repeated until the engine **19** stops.
(65) Note that when a plurality of temperature sensors **50** are provided such that the temperature sensor **50** is disposed in each of the plurality of radar devices **40**, for example, in step S**20**, it is determined whether or not the temperature detected by any one of the temperature sensors **50** is equal to or higher than the first threshold, and when even one detected temperature exceeds the first threshold, the control proceeds to step S**30**. Also, in step S**40**, it is determined whether or not all the detected temperatures are equal to or lower than the second threshold, and when all the detected temperatures are equal to or lower than the second threshold, control proceeds to step S**50**.
(66) As described above, when the temperature of the radar device **40** rises, the fan **31** is changed to the intake state from the exhaust state by rotating the fan **31** in the reverse direction. Thereby the air from the outside hits the radar device **40** and it is possible to cool the radar device **40** and to lower the temperature of the radar device **40**. Further, when the temperature of the radar device **40** is lowered by air sucked from the outside, the radiator **20** can be cooled by returning to the exhausted state.
Embodiment 2
(67) Next, the wheel loader **100** of the second embodiment according to the present disclosure will be described.
(68) In the wheel loader **100** of the first embodiment, the radiator **20** is cooled in the exhaust state, but in the wheel loader **100** of the second embodiment, the radiator **20** is cooled in the intake state.
(69) FIG. **6** is a side view illustrating the rear part of the wheel loader **100** of the second embodiment. As shown in FIG. **6**, in the wheel loader **100** of the second embodiment, the engine **19**, the fan **31**, and the radiator **20** are disposed in order from the front along the front-rear direction X, unlike the order in the first embodiment. With such a configuration, the radiator **20** can be cooled in the intake state (see arrow C).
(70) FIG. **7** is a diagram illustrating the configurations of the cooling device **30** and the control of wheel loader **100** of the second embodiment. In the second embodiment, the fan **31** is installed so

that the fan **31** enters the intake state by driving the hydraulic pump **32** to rotate fan **31** in the state in which the directional control valve **37** is set to the first position P**1**. Therefore, when the directional control valve **37** is set to the second position P**2**, the fan **31** enters the exhaust state by driving the hydraulic pump **32** to rotate the fan **31**.

(71) As shown in FIG. **7**, the wheel loader **100** of the second embodiment includes an imaging device **70** instead of the temperature sensor **50**. The imaging device **70** captures the surface state of the radar device **40**. The imaging device **70** particularly captures the surface state of the surface (the surface from which the millimeter wave radar is emitted and incident) of the cover part **41***b* facing the rear of the radar device **40**. The captured image data is transmitted to the controller **60**. The controller **60** normally sets the directional control valve **37** to the first position P**1** and sets the fan **31** to the intake state, and the controller **60** switches the directional control valve **37** to the second position P**2** when determining that the surface of the radar device **40** is dirty by image analysis of the captured image data.

(72) As a result, dirt such as dust adhering to the radar device **40** can be blown off and dirt can be reduced

(73) (Action)

(74) Next, control of the wheel loader **100** of the second embodiment will be described. FIG. **8** is a flowchart illustrating control of wheel loader **100** of the present embodiment.

(75) First, in step S**110**, when the engine **19** of the wheel loader **100** is started, the controller **60** sets the directional control valve **37** to a state in which the valve body is disposed at the first position P**1**, drives the hydraulic pump **32** to rotate the fan **31** in the normal direction, and sets the fan to the intake state.

(76) Next, in step S**120**, the controller **60** analyzes the image data captured by the imaging device **70** and determines whether or not the surface of the radar device **40** is dirty. Here, as a determination of whether or not it is dirty, for example, the case that the current image data is compared with the initial state and the surface gradation of the radar device **40** is darker than a predetermined gradation can be mentioned. Since the gradation can be obtained as a numerical value by, for example, black-and-white gradation of the image data, the predetermined gradation can also be set as a numerical value.

(77) When it is determined that the surface of the radar device **40** is not dirty, the control of step S**120** is repeated, and the fan **31** rotates in the normal direction to maintain the intake state.

(78) On the other hand, when it is determined in step S**120** that the surface of the radar device **40** is dirty, in step S**130** the controller **60** switches the directional control valve **37** from the first position P**1** to the second position P**2** to rotate the fan **31** in the reverse rotation and sets the fan **31** to the exhaust state.

(79) Next, in step S**140**, the controller **60** determines whether or not the dirt has been removed based on the image data detected by the imaging device **70**. Here, the threshold value for determining whether or not the dirt has been removed in step S**140** can be set to a value (closer to white) lower than the threshold value in step S**120**. When it is determined that the dirt has not been removed, the control of step S**140** is repeated to maintain the exhaust state in which the fan **31** rotates in the reverse direction.

(80) On the other hand, when it is determined in step S**140** that the dirt has been removed, in step S**150** the controller **60** switches the directional control valve **37** from the second position P**2** to the first position P**1** to rotate the fan **31** in the normal direction and sets the fan **31** to the intake state. After step S**150**, the control returns to step S**120**, and steps S**120** to S**150** are repeated until the engine **19** stops.

(81) In the case that a plurality of radar devices **40** are provided, for example, when it is determined in step S**120** that the surface of any one radar device is dirty, control proceeds to step S**130**. When it is determined in step S**140** that dirt has been removed from the surfaces of all radar devices **40**, control proceeds to step S**150**.

(82) As described above, when the surface of the radar device **40** becomes dirty, the fan **31** is changed to the exhaust state from the intake state by rotating the fan **31** in the reverse direction. Thereby the air from the inside of the engine room **6** can be hit to the radar device **40** and it is possible to blow off dirt from the radar device **40**. Further, when the dirt is removed, the fan **31** is returned to the intake state from the exhaust state, and the radiator **20** can be cooled.
(83) (Characteristics)
(84) (1)
(85) The wheel loader **100** (an example of a work machine) of each of the first embodiment and the second embodiment include the vehicle body **1** that is able to travel, the fan **31**, the radar device **40** (an example of an obstacle detection device), and a controller **60**. The fan **31** exchanges air between the inside and the outside of the vehicle body **1** through the opening **6***c* provided toward the rear of the vehicle body **1**. The radar device **40** is installed behind the fan **31** and detects an obstacle S in the rear of the vehicle body **1**. The controller **60** controls the intake and exhaust of the fan **31** based on the state of the radar device **40**.
(86) As a result, for example, in the case of cooling normally in the exhaust state as in the first embodiment, the fan **31** is set to the intake state to reduce the temperature of the radar device **40** when the temperature of the radar device **40** rises. On the other hand, in the case of cooling normally in the intake state as in the second embodiment, when the dirt on the radar device **40** becomes heavy, the dirt on the radar device **40** can be removed by setting the fan **31** to the exhaust state.
(87) Therefore, it is possible to suppress deterioration in the performance of the radar device **40** detecting an obstacle in the rear.
(88) (2)
(89) In wheel loader **100** (an example of a work machine) of the first embodiment, the fan **31** cools the inside of vehicle body **1** in the exhaust state in which air is exhausted from the inside of the vehicle body **1** to the outside.
(90) As a result, when the temperature of the radar device **40** rises due to exhaust from the inside of the vehicle body **1** to the outside, the temperature of the radar device **40** can be lowered by setting the fan **31** to the intake state.
(91) (3)
(92) The wheel loader **100** (an example of a work machine) of the first embodiment further includes the temperature sensor **50**. The temperature sensor **50** detects the temperature of the radar device **40**. When the detected value of the temperature sensor **50** is equal to or higher than a predetermined temperature, the controller **60** controls the fan **31** so as to set from the exhaust state to the intake state in which air is sucked from the outside to the inside of the vehicle body **1**.
(93) As a result, it is possible to detect that the temperature of the radar device **40** has risen due to the exhaust from the inside of the vehicle body **1** to the outside, so that the temperature of the radar device **40** can be lowered by setting the fan **31** to the intake state.
(94) (4)
(95) In wheel loader **100** (an example of a work machine) of the second embodiment, the fan **31** cools the inside of vehicle body **1** in the intake state in which air is sucked from the outside of vehicle body **1** toward the inside.
(96) As a result, when the radar device **40** becomes dirty due to the intake from the outside to the inside of the vehicle body **1**, the fan **31** is set to the exhaust state to blow away the dirt and reduce the dirt on the radar device **40**.
(97) (5)
(98) The wheel loader **100** (an example of a work machine) of the second embodiment further includes the imaging device **70** (an example of a dirt detector). The imaging device **70** detects dirt on the radar device **40**. When the controller **60** determines that the radar device **40** is dirty based on the detection by the imaging device **70**, the controller **60** controls the fan **31** so that air is exhausted

from the inside of the vehicle body **1** to the outside.
(99) As a result, it is possible to detect that the radar device **40** is dirty due to the intake from the outside to the inside of the vehicle body **1**, so that the dirt of the radar device **40** can be reduced by setting the fan **31** to the exhaust state and blowing off the dirt.
(100) (6)
(101) In wheel loader **100** (an example of a work machine) of the second embodiment, the imaging device **70** (an example of an imaging section) captures the surface state of the radar device **40**. The controller **60** determines dirt on the radar device **40** based on the image captured by the imaging device **70**.
(102) As a result, dirt on the radar device **40** can be detected based on the image captured by the imaging device **70**.
(103) (7)
(104) The wheel loader **100** (an example of a work machine) of the first embodiment further includes the engine **19** and the radiator **20**. The engine **19** is disposed inside the vehicle body **1**. The radiator **20** is disposed between the fan **31** and the engine **19**.
(105) As a result, the inside of the vehicle body **1** is cooled in an exhaust state in which the air is exhausted from the inside of the vehicle body **1** toward the outside, and when the temperature of the radar device **40** rises, the temperature of the radar device **40** can be lowered by setting the fan **31** to the intake state.
(106) (8)
(107) In the wheel loader **100** (an example of a work machine) of each of the first embodiment and the second embodiment, the vehicle body **1** includes the vehicle body frame **2** and the work implement **3** attached to the front of the vehicle body frame **2**.
(108) Thereby, in the wheel loader **100** in which the work implement **3** is provided in front, it is possible to suppress deterioration in the performance of the radar device **40** that detects an obstacle in the rear.
(109) (9)
(110) The wheel loader **100** (an example of a work machine) of each of the first embodiment and the second embodiment includes the hydraulic pump **32** and the hydraulic motor **34**. The hydraulic pump **32** is driven by the shaft output of the engine **19**. The hydraulic motor **34** is driven by hydraulic fluid supplied from the hydraulic pump **32**. The fan **31** is driven by the shaft rotation output of the hydraulic motor **34**.
(111) As a result, the fan **31** can be rotated by hydraulic drive.
OTHER EMBODIMENTS
(112) Although an embodiment of the present invention has been described so far, the present invention is not limited to the above embodiment and various modifications may be made within the scope of the invention.
(113) (A)
(114) While in the first embodiment, the fan **31** is automatically controlled from the exhaust state to the intake state based on the detected value of the temperature sensor **50**, the present invention is not limited to this. For example, as shown in FIG. **9**, an operation switch **150** (an example of an operation section) for switching the rotation of the fan **31** between the normal rotation and the reverse rotation and a display section **151** for displaying the temperature detected by the temperature sensor **50** may be provided. In this case, the operator may check the display section **151** and press the operation switch **150** so as to properly set the fan **31** to the intake state based on the temperature displayed on the display section **151**.
(115) Accordingly, by the operator operating the operation switch **150** based on the temperature displayed on the display section **151**, the fan **31** is set to the intake state and the temperature of the radar device **40** can be lowered. Also, the operator can return to the exhaust state from the intake state based on the displayed temperature.

(116) (B)
(117) While in the above-described second embodiment, the fan **31** is automatically controlled from the intake state to the exhaust state based on the image data of the imaging device **70**, the present invention is not limited to this. For example, as shown in FIG. **10**, an operation switch **170** (an example of an operation section) for switching the rotation of the fan **31** between the normal rotation and the reverse rotation, and a display section **171** for displaying an image captured by the imaging device **70** may be provided. In this case, the operator may check the display section **171** and press the operation switch **170** so as to properly set the fan **31** to the exhaust state based on the image on the display section **171**.
(118) Accordingly, by the operator operating the operation switch **170** based on the image displayed on the display section **151**, the fan **31** is set to the exhaust state and the dirt can be blown off from the radar device **40**. In addition, the operator can return to the intake state when determining that the dirt has been removed based on the image.
(119) (C)
(120) While in the first embodiment, the fan **31** is automatically controlled from the exhaust state to the intake state based on the detected value of the temperature sensor **50**, the temperature sensor **50** may not be provided. For example, the fan **31** may be automatically controlled so as to be in the intake state for a preset period at predetermined intervals during the operation of the engine **19**.
(121) In this manner, the temperature of the radar device **40** can be lowered by the controller **60** controlling the fan **31** from the exhaust state to the intake state at predetermined time intervals.
(122) (D)
(123) While in the above-described second embodiment, the fan **31** is automatically controlled from the intake state to the exhaust state based on the image data of the imaging device **70**, the imaging device **70** may not be provided. For example, the fan **31** may be automatically controlled so as to be in the exhaust state for a preset period at predetermined intervals during the operating of the engine **19**.
(124) In this manner, dirt can be removed from the radar device **40** by the controller **60** controlling the fan **31** from the intake state to the exhaust state at predetermined time intervals.
(125) (E)
(126) While in the above-described second embodiment, dirt is detected using the imaging device **70** as an example of the dirt detector, but the dirt detector is not limited to this. For example, dirt on the surface of the radar device **40** may be detected using a reflective laser sensor or the like.
(127) (F)
(128) The arrangement of the radar devices **40** is not limited to the configuration of the above-described embodiments. One or a plurality of radar devices **40** may be disposed above or below one radar device **40** provided on the left side of the central axis A. Further, one or a plurality of radar devices **40** may be disposed above or below one radar device **40** provided on the right side of the central axis A.
(129) (G)
(130) In the above-described embodiments, the plurality of radar devices **40** are disposed symmetrically with respect to the central axis A, but they may not be disposed symmetrically.
(131) (H)
(132) While the radar devices **40** are fixed to the grille **6***b* in the above embodiments, the present invention is not limited to this. The radar device **40** may be attached to the counterweight **10**, for example. When the radar device **40** is attached to the counterweight **10**, the radar device **40** is not limited to being attached to the surface of the counterweight **10**, and may be embedded.
(133) (I)
(134) While the wheel loader is used in the discussion as an example of a work machine in the above embodiments, the present invention is not limited thereto and the work machine may be a dump truck, motor grader, hydraulic excavator, forklift, or the like.

(135) According to the work machine and the control method for the work machine of the present disclosure, it is possible to suppress deterioration in the performance of detecting an obstacle in the rear, and is useful as a wheel loader or the like.

Claims

1. A work machine comprising: a vehicle body configured to travel; a fan configured to exchange air between an inside and an outside of the vehicle body through an opening provided toward a rear of the vehicle body; an obstacle detection device installed behind the fan, the obstacle detection device being configured to detect an obstacle in the rear of the vehicle body; a controller configured to control intake and exhaust of the fan based on a state of the obstacle detection device; and a temperature sensor configured to detect a temperature of the obstacle detection device, the fan being configured to cool the inside of the vehicle body in an exhaust state in which air is exhausted from the inside of the vehicle body toward the outside, and the controller being further configured to control the fan so as to set from the exhaust state to an intake state in which air is sucked from the outside to the inside of the vehicle bod when a value detected by the temperature sensor is equal to or higher than a predetermined temperature.
2. The work machine according to claim 1, further comprising: an engine disposed the inside of the vehicle body, and a radiator disposed between the fan and the engine.
3. The work machine according to claim 1, wherein the vehicle body includes a vehicle body frame and a work implement attached in front of the vehicle body frame.
4. The work machine according to claim 1, further comprising: a pump driven by a shaft output of the engine, and a motor driven by hydraulic fluid supplied from the pump, the fan being driven by a shaft rotation output of the motor.
5. The work machine according to claim 1, further comprising: an operation section configured to switch the fan from the exhaust state to an intake state in which air is sucked from the outside of the vehicle body toward the inside through the opening, the controller being configured to control the fan so as to set from the exhaust state to an intake state in which air is sucked from the outside of the vehicle body toward the inside based on an operation of the operation section.
6. The work machine according to claim 1, wherein the controller is configured to control the fan so as to set from the exhaust state to an intake state in which air is sucked from the outside of the vehicle body toward the inside at predetermined time intervals.
7. A work machine comprising: a vehicle body configured to travel; a fan configured to exchange air between an inside and an outside of the vehicle body through an opening provided toward a rear of the vehicle body; an obstacle detection device installed behind the fan, the obstacle detection device being configured to detect an obstacle in the rear of the vehicle body; a controller configured to control intake and exhaust of the fan based on a state of the obstacle detection device; and a dirt detector configured to detect dirt on the obstacle detection device, the fan being configured to cool the inside of the vehicle body in an intake state in which air is sucked from the outside of the vehicle body toward the inside, and the controller being further configured to control the fan so as to set from the intake state to an exhaust state in which air is exhausted from the inside of the vehicle body toward the outside when the controller determines that the obstacle detection device is dirty based on a detection by the dirt detector.
8. The work machine according to claim 7, further comprising: an engine disposed in the vehicle body; and a radiator disposed between the fan and the engine.
9. The work machine according to claim 7, wherein the vehicle body includes a vehicle body frame and a work implement attached in front of the vehicle body frame.
10. The work machine according to claim 7, further comprising: a pump driven by a shaft output of the engine, and a motor driven by hydraulic fluid supplied from the pump, the fan being driven by a shaft rotation output of the motor.

11. The work machine according to claim 7, wherein the dirt detector is an imaging device configured to capture a surface state of the obstacle detection device, and the controller is configured to determine dirt on the obstacle detection device based on an image captured by the imaging device.